HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and implemented **Python** scripts to automate data extraction and transformation processes, reducing manual effort by approximately **40%** for batch reporting.
- Designed and built **analytical tools** using **Python** for energy modelling studies, focusing on efficiency measures and heating system decarbonisation options.
- Prepared, integrated, and analysed complex datasets to refine inputs and outputs for energy modelling simulations, ensuring data integrity and improving accuracy.
- Authored comprehensive reports and generated insightful **data visualisations** to effectively communicate findings from energy modelling studies to stakeholders.
- Researched and identified current and future **data gaps** within energy datasets, continuously developing methodologies and enhancing analytical tool functionality.
- Collaborated with cross-functional teams to gather requirements for new analytical tools and integrate them into existing energy sector workflows.
- Managed and supported production environments for enterprise applications, ensuring adherence to strict SLAs and performing incident response for P1/P2 issues using **Splunk** and **Dynatrace**.

SKILLS

Languages – Python, SQL, R

Tools & Technologies – Power BI, Tableau, Pandas, NumPy, Scikit-learn, Flask, FastAPI, Splunk, Dynatrace, Jenkins, Git, Docker, Jira, Confluence

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle

Methodologies – Agile, Scrum

Domain – Energy Modelling, Data Analysis, Data Visualisation, Business Intelligence, Production Support, ETL

PROJECTS

Energy Efficiency Modelling Tool

- Developed a **Python**-based application to model the impact of various energy efficiency measures on commercial buildings, processing data for ~50 hypothetical building profiles.
- Integrated **SQL** queries to extract historical energy consumption data from a simulated database, feeding into the modelling engine.
- Created interactive **data visualisations** using Matplotlib and Seaborn to present model outputs, highlighting potential savings and ROI for stakeholders.

Heating System Decarbonisation Analysis

- Built a **Python** script to analyse different heating system decarbonisation options (e.g., heat pumps, biomass) for residential properties, assessing feasibility based on ~20 input parameters.
- Performed data cleaning and transformation using **Pandas** on diverse datasets, including property characteristics and regional energy prices.
- Researched and documented emerging data requirements for future iterations of the decarbonisation analysis tool.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)